Linear Logic: Resources, Proof Reduction, and Feedback

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Linear logic restricts the invisible copying and discarding of hypotheses that ordinary propositional reasoning permits. This becomes useful when a premise represents a consumable resource. A proof can then be read as a resource-transforming program, and cut elimination as the composition of such programs. The Geometry of Interaction develops this dynamical idea through feedback operators and paths in networks.[1]

§1.1 The tensor rule cannot spend one resource twice

In the intuitionistic linear sequent calculus, the rule for tensor splits the available context:

$$\frac{\Gamma \vdash A \qquad \Delta \vdash B}{\Gamma, \Delta \vdash A \otimes B}.$$

A hypothesis allocated to the left premise cannot also be allocated to the right one. Hence there is no general derivation $A \vdash A \otimes A$. The missing step is contraction, the ordinary rule that duplicates an assumption. Weakening, which discards an unused assumption, is also unavailable for a general linear hypothesis.

The exponential modality $!A$ permits controlled reuse. A linear implication $A \multimap B$ represents a transformation consuming an A and producing a B; a reusable transformation is supplied as $!(A \multimap B)$. In a symmetric monoidal closed model, the formula is not just suggestive notation:

$$\mathrm{Hom}(A \otimes B, C) \cong \mathrm{Hom}(A, B \multimap C).$$

Linear implication is the internal function object adjoint to tensoring with an input resource.

§1.2 A small story is an exact transition system

Let P, S, T, B, W represent a pig, straw, sticks, bricks and wolf, and H_S, H_T, H_B the completed houses. Start with

$$M_0 = 3P + S + T + B + W,$$

where addition records multiplicity in a multiset. Consider the six rewrite rules

$$\begin{aligned} P \otimes S &\multimap H_S, & P \otimes T &\multimap H_T, & P \otimes B &\multimap H_B, \\ W \otimes H_S &\multimap W, & W \otimes H_T &\multimap W, & W \otimes H_B &\multimap H_B. \end{aligned}$$

After using each construction rule once, the state is $W + H_S + H_T + H_B$. Applying the three wolf rules gives the completely explicit trace

$$\begin{aligned} W + H_S + H_T + H_B &\longmapsto W + H_T + H_B \\ &\longmapsto W + H_B \\ &\longmapsto H_B. \end{aligned}$$

[1] See J.-Y. Girard, *Linear Logic* (1987), and T. Seiller, *Interaction Graphs: Additives* for a mathematical feedback interpretation.

No pig or material remains: resources have been consumed, while the rules themselves may be treated as reusable assumptions.

The same computation is a Petri-net transition. Record multiplicities by a vector $m \in \mathbb{N}^n$. A rule with input vector a and output vector b can fire exactly when $m \geq a$ coordinatewise, and then

$$\boxed{m \longmapsto m - a + b.}$$

This tests the resource accounting in a way that a narrative analogy cannot. It also distinguishes the persistent availability of a rule from the transient availability of the tokens to which that rule applies.

§1.3 Additive choice is not simultaneous consumption

Tensor $A \otimes B$ supplies both resources, while the additive conjunction $A \,\&\, B$ means that the environment can select either branch. The two right introduction rules have different resource behavior:

$$\frac{\Gamma \vdash A \qquad \Delta \vdash B}{\Gamma, \Delta \vdash A \otimes B}, \qquad \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \,\&\, B}.$$

The shared context in the second rule does not license simultaneous consumption twice. The premises describe alternative interactions; the opponent selects which one occurs. Additive disjunction $A \oplus B$ lets the prover instead select a branch. A protocol offering either of two services on demand is therefore different from one physically providing both outputs.

§1.4 Cut elimination composes resource transformers

The cut rule composes a proof producing A with another proof consuming A:

$$\frac{\Gamma \vdash A \qquad \Delta, A \vdash B}{\Gamma, \Delta \vdash B}.$$

Under proofs-as-programs, a principal cut can reduce like a linear beta step

$$(\lambda x.t)\, u \longrightarrow t[u/x],$$

where the bound variable x occurs exactly once in the linear term t. The argument u is therefore neither silently duplicated nor erased.

The first construction and demolition in our resource trace compose into a direct rule $P \otimes S \otimes W \multimap W$. The intermediate object H_S disappears because its producer has been connected to its consumer. A cut-free composite describes the overall transformation without keeping that internal communication visible.

§1.5 The feedback formula counts internal paths

Let a linear process accept an external input x and an internal input v, and return an external output y and an internal output v':

$$\begin{pmatrix} y \\ v' \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x \\ v \end{pmatrix}.$$

Feedback identifies $v' = v$, so $v = Cx + Dv$. If $I - D$ is invertible, eliminating this hidden state gives

$$\boxed{y = (A + B(I - D)^{-1}C)x.}$$

For nilpotent $D^r = 0$,

$$A + B(I - D)^{-1}C = A + BC + BDC + \cdots + BD^{r-1}C.$$

Each summand records paths passing through the internal connection a specified number of times.

Choose

$$A = 0, \quad B = (1 \;\; 0), \quad C = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Here $D^2 = 0$, $BC = 0$, but $BDC = 1$. Thus the external map is the identity,

$$A + B(I - D)^{-1}C = 1.$$

The input reaches the output only after circulating through the internal connection. The block A alone would entirely miss this effect.

Girard's Geometry of Interaction uses operator-level execution formulas to model cut elimination. This finite-dimensional feedback calculation illustrates the algebraic mechanism; a full logical semantics also requires appropriate typing and proof-correctness conditions. The connection is substantive because eliminating an internal wire is literally an inverse or sum of path operators.

$La \longrightarrow L(^*m) \otimes L((Rm) \parr a)$

may be depicted in the following way:

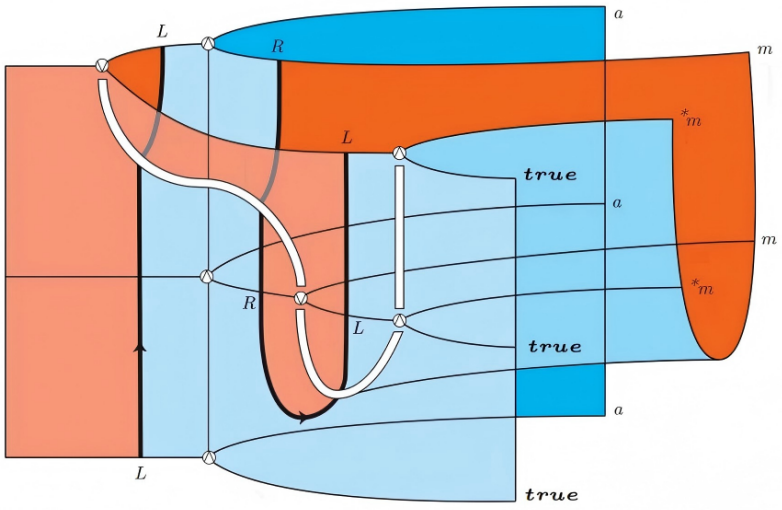


Figure 1.1: Wired inputs and outputs visualize proof composition as controlled interaction.

§1.6 Negation as testing against possible opponents

In Ludics a design can be tested against another design. Write $D \perp E$ if their specified interaction terminates successfully, and for a family $\mathcal{A}$ define

$$\mathcal{A}^{\perp} = \{E : D \perp E \text{ for every } D \in \mathcal{A}\}.$$

The double orthogonal satisfies the closure laws

$$\mathcal{A} \subseteq \mathcal{A}^{\perp\perp}, \qquad \mathcal{A}^{\perp\perp\perp} = \mathcal{A}^{\perp}.$$

These follow from the elementary reversal of inclusion: $\mathcal{A} \subseteq \mathcal{B} \Rightarrow \mathcal{B}^{\perp} \subseteq \mathcal{A}^{\perp}$. Families fixed by double orthogonality are called behaviors.

A linear proof consumes precisely specified resources, cut elimination composes its transformations, feedback semantics describes the internal paths of interaction, and orthogonality characterizes a behavior by the opponents against which it succeeds. The common structure is an explicit operation on processes, rather than a string of analogies about stories or games.